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RESEARCH-ARTICLE

Making recruitment more inclusive: unfairness monitoring with a job matching machine-learning algorithm

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Making Recruitment More Inclusive: Unfairness Monitoring With A Job Matching Machine-Learning Algorithm

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ABSTRACT

For decades human resources management has relied on recruitment processes rooted in self-reports (such as surveys, questionnaires, personality and cognitive tests) and interviews that were, for most of them, lacking scientific rigor and replicability. Here, we introduce an algorithm that matches job offers and workers that not only outperforms the classic recruitment and job matching methods, but that has at its core algorithmic safeguards to prevent (as much as possible) the pitfall of unfairness and discrimination. Our approach to algorithm development is guided by the constant goal of offering a solution at the cutting edge of technology that has at its core a strict policy of being as fair, inclusive and transparent as possible.

KEYWORDS

Social Impact Software, Fairness, Inclusion, Recruitment, Equal Opportunity

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1 INTRODUCTION

Technology should be, first and foremost, aiming at improving people's lives, at the individual and collective levels. In other words,

every stakeholder involved in a scientific and technological discovery, whether aimed at being commercialized or not, should bear in mind that its outcomes should not elicit, directly or not, voluntarily or not, any form of harm, inequity, exclusion, unfairness or discrimination. But that is not all. A technological innovation should not be solely aiming at avoiding negative outcomes. Above and beyond, it should be designed (and deployed) for positive social impact purposes, i.e., with the intent of benefiting individuals and society at a large [56].

One could consider what is often referred to as Artificial Intelligence (AI) as the ultimate advance in science and technology to date. We often read that for AI to impact society positively, it should *a minima* be objective, neutral, and unbiased. These are very noble objectives, and chances are for them to be embraced by a great majority of people. Should these goals be realistic [33]. But there is a well-known and extensively reported problem with objectivity, neutrality and unbiasedness: They simply do not exist from an operational point of view. The reason is they are impossible to implement in the context of AI – or in any other field or topic for that matter – as long as human beings are involved, one way or another [12].

One of the biggest misconceptions about AI, or should we say, one of the mainstream expectation and fear, is for it to do things better than a human. In a lot of cases, it does. But “better” should not be the main goal. AI-powered solutions should be developed to do things differently [48]. Humans make mistakes and, as part of the social contract over centuries, other humans have monitored, corrected, anticipated and sometimes punished those (who made) mistakes. There is also a form of understanding, if not tolerance for some of these mistakes at the societal level. For a significant portion of them, algorithms are developed by humans, or by other algorithms, that at one point were developed by humans. Yet, algorithms are not humans. Hence, our tolerance for algorithmic mistakes should be lower than our tolerance for human error [4].



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The human factor comes with greatness and flaws such as biases. Yet, as addressed in some of our previous work [35] it is a profound misconception to consider human biases an exception to a norm rooted in (the economic definition of) rationality. Biases are everywhere, whether we like it or not they are the norm more than they are the exception. It is neither a good nor a bad thing. It just happens to be the way human beings behave and make decisions despite what supporters of a rational approach to economic theory have been disseminating for over a century [26]. This means that data collected on humans and their behaviors are intrinsically biased. Hence, in addition to addressing the issue of biased algorithms and their possible unfair outcomes, it is necessary to address the issues related to biased datasets [32].

It is therefore our duty, as a technology company developing human resources (HR) solutions powered by AI, as for any other company leveraging or developing systems with AI at their core, to not only be aware of these issues, but to make sure that we build the right regulatory framework — enforced by informed regulatory bodies and policymakers as well as by companies and individuals themselves — and the right technology for AI solutions to be (as) inclusive and fair (as technology allows).

What AI, like any other innovation and technology that aims at having positive social impact, should be aspiring to is therefore to provide the most rigorous and transparent methodology for an expected outcome to be reached. By minimizing, as much as possible, the negative impact of biases, regardless of them coming from datasets, algorithms, and/or designers, coders, and developers. This should be made possible, and available to all individuals, thanks to a set of principles coupled with actionable processes and guidelines in compliance to existing regulatory measures and upcoming ones, if or when necessary. Above and beyond, we firmly believe it should be mandatory to put inclusion, fairness and human wellness and safety at the core of the design, implementation and deployment at scale of any technological innovation for it to be deemed responsible. This should be the case with AI-powered Human Resources Management (HRM) solutions [49], i.e., artificial intelligence created and used to automate HR tasks such as job matching.

In human resources, and more particularly during recruitment processes, whether for the lack of anything better or because of the impossibility to implement more rigorous solutions at scale, public and private organizations still rely heavily on declarative methods (e.g., self-reported data collected thanks to surveys, questionnaires or tests) despite more than forty years of solid scientific literature highlighting their lack of reliability, reproducibility, and predictive power of job fit and/or performance [54]. For nearly fifty years, the professional careers (and by extension the professional and personal lives) of millions of individuals have been shaped by the results of these unfair methods.

Our company was founded with the goal that AI could not only accelerate job recruitment processes (especially the part that matches candidates with job offers) but also minimize the unfairness of current mainstream job recruitment and matching processes. In order to achieve this, decisions made by our algorithms must be methodologically and scientifically rigorous as well as evidence-informed. Our R&D effort was not solely focused on algorithms

to improve matching performance, but also to develop companion algorithms that act as safeguards to minimize unfairness and discrimination.

When a human resources process, regardless its level of automation, does a poor job at matching or recruiting, the consequences can be dramatic for the candidates turned employees (or not). They can experience distress, mental health issues and bad job performances [15]. Another consequence of mismatching a candidate and a job not to be neglected is the increased financial costs for the employer [16]. This is simply not acceptable.

As a temporary recruitment agency leveraging AI to optimize its job matching process and make it as fair as possible, Gojob developed a job matching machine learning solution consisting in a scoring algorithm that is able to identify the most relevant temporary workers for a given client request (i.e., a job offer). Our algorithm is a tool for internal recruiters to help them staff specific HR needs as fast and as accurately as possible. Four principles are at the core of the research and development effort: (1) to perform the most relevant and responsible recommendations to match a specific job description; (2) to do it in the shortest amount of time; to (3) make sure that the algorithm processes the most up-to-date dataset and finally (4) for all previous three steps to be conducted in a fair and inclusive way. This is why, the main goal we have is to give every worker/candidate in our database equal chances to be recruited regardless their age, gender, origin, as long as they exhibit the best fit with the job offer in terms of skills, professional experience and education. In our approach, the key features should be the worker's professional experiences and motivation, the latter being assessed thanks to a combination of natural language processing and cognitive features embedded in our algorithm. Therefore, it is crucial to check whether our mathematical models make decisions and predictions based on the relevance and the quality of an application and not on other criteria that may be correlated or confounded. An algorithm should not only be considered in light of its performance and outcomes according to a given set of metrics, but also according to the context in which its outcomes (i.e., the decision it makes) are being used.

2 TOWARDS RESPONSIBLE HUMAN RESOURCE MANAGEMENT TECHNOLOGY

In human resources management, and more particularly in recruitment and job matching processes, the “technology” that has been the most popular and widely used over the past decades by organizations is rooted in questionnaires, surveys, tests and interviews [29]. Such declarative methods are used with the intent to capture someone's personality traits, cognitive abilities and skills, but also to provide predictive insights regarding job fit as well as success or failure in job performance. In other words, a series of questions is expected to unveil the complexity of a candidate's behavior, personality, affective reactions, skills and brain functioning. But there is a major problem. The reliability and predictive power of recruitment tests has been extensively challenged since the 1970s when the first peer-reviewed articles were published demonstrating the limitations and flaws of questionnaires and self-reports [31].

Unlike psychological states that evolve over time, traits (that some HRM tests claim they can capture) should not sensitive to

context. This means that any methodology meant to measure personality traits that would not produce the same outcomes under various contexts, or at different points in time, should not be considered fair.

As artificial intelligence became pervasive, legit concerns started to grow at the societal level regarding the so-called “black box” functioning of AI-powered solutions developed for commercial purposes. Some of these concerns are justified when the output of algorithms exhibits obvious biases [47] that lead to unfairness in the way people are benefiting from AI-powered technology. Or worse, when the decisions [42] (informed or taken) by AI algorithms lead to obvious discrimination [14, 27] or mistakes that cannot be identified even when second opinions are required to correct them [6]. This issue is key in many business sectors including in human resources management where AI’s role is getting more and more prevalent, and some automated decisions made by algorithms can have dramatic consequences on someone’s professional career.

Requiring algorithmic transparency is necessary but hard to implement. One of the reasons is the clear tension between businesses wanting to protect intellectual property that required millions of dollars invested in research and development and the moral and legal need for transparency and accountability [9, 25].

A significant amount of businesses has taken proactive measures to not only introduce principles and guidelines [18, 37, 39] when it comes to using AI, but to make them operational (e.g., Microsoft company¹). Although there is no real consensus regarding the way these principles are presented or implemented, they should involve epistemic principles and more general ethical principles [9]. Yet, although companies commit (genuinely or for public relation risk mitigation purposes) to responsible AI principles, there is not yet a standard guiding the road from principles to implementation and regulation.

Interestingly, and luckily, responsible AI has become a business within the business. Professional services firms offer to assist companies in their implementation of responsible AI principles [28], and specific tech solutions are now available to assist companies, such as Credo AI² offering “an end-to-end governance platform for managing compliance and measuring risk for your AI deployments at scale”.

It is noteworthy that, because they are under constant public opinion, media and regulatory scrutiny, big tech companies have made significant (and visible) efforts to adopt and most of the time implement responsible AI principles. These initiative, although gaining momentum, are not generalized yet in the startup ecosystem but are slowly but surely becoming the norm. Several small businesses do their best to try to fly under the radar when it comes to responsible AI, or assessment of the reality of their AI solutions. There is for instance a so-called “fake AI” issue: Companies claiming to develop AI solutions when they are not [45]. This is not just an unsustainable stance from both a business standpoint, it is also a major ethical and regulatory issue.

In April 2021, the European Commission (EC), issued the proposal 2021/0106(COD) entitled “Laying down harmonized rules on Artificial Intelligence (Artificial Intelligence Act) and amending

certain union legislative acts” [11]. What is referred to as “EC’s Artificial Intelligence Act” was issued to:

- (1) Ensure that AI systems placed on the Union market and used are safe and respect existing law on fundamental rights and Union values;
- (2) Ensure legal certainty to facilitate investment and innovation in AI;
- (3) Enhance governance and effective enforcement of existing law on fundamental rights and safety requirements applicable to AI systems;
- (4) Facilitate the development of a single market for lawful, safe and trustworthy AI applications and prevent market fragmentation.

Amongst the many points raised, transparency in “certain” AI systems are highlighted, as well guidelines for governance and implementation, while not preventing innovation. The EC effort is to be supported, and seems to gain momentum abroad. For instance, in November 2010, the US Director of the National AI Initiative at the White House Office of Science and Technology Policy, as she was speaking at an event organized by Fortune Magazine, hinted at the US having to model Europe’s approach to AI regulation [43]. In the United States, as of 2021, artificial intelligence bills were introduced in seventeenth states and enacted in four of them [30].

3 A FAIRNESS AND INCLUSIVE FRAMEWORK FOR HUMAN RESOURCES TECH

Although the work focusing on defining and studying fairness with a multidisciplinary approach dates back to the 1960s [21], the interest in machine learning research in this area has only really boomed in the mid 2010s [10]. As Raghavan et al. [40] very accurately stressed, it can be difficult for an external entity to study the fairness of the models used by a company, and the often sensitive and proprietary data that the models are crunching. It is therefore key for the effort on fairness to be embedded as soon as the first mathematical models are created.

In their study, van den Broek et al. [51] acknowledge how complex it is to take fairness issues into account in recruitment processes but they also offer interesting guidelines to considering when one wants to build responsible and fair AI.

In light of the diversity of definitions and the context-dependence of fairness judgments, and despite some challenges when trying to automate fairness [53], we have therefore decided to introduce in previous work our own framework to help define what we think fair technology should be. We do believe that fair HR technology is an innovation that, at the moment in time when it is being introduced to the market, must be:

- (1) **LEGAL** – It must comply to the general law as well as regulations that are specific to the sector of activity and to the technology developed. Lawfulness and compliance are mandatory.
- (2) **ETHICAL & SAFE** – Not harming in any way its users and its non-users.
- (3) **PROVABLE & SCIENTIFICALLY VALIDATED** – Technology, if not validated by rigorous science, cannot be proved nor can it be explained.

¹<https://www.microsoft.com/en-us/ai/our-approach>

²<https://www.credo.ai>

- (4) **ROBUST & RELIABLE** – The outcomes of the tech should be replicable.
- (5) **NON-DISCRIMINATORY** – The outcomes of the tech should not vary depending on who benefits from them
- (6) **CUTTING EDGE** – The inventors of the tech solution must be convinced that there is no better solution to achieve the desired outcomes when it is being release.
- (7) **HAVE REALISTIC AND MEASURABLE GOALS** – Aiming at the impossible but achieving the possible.
- (8) **INCLUSIVE & ACCESSIBLE** – Everyone should have equal access to innovation.
- (9) **PURPOSEFUL** – By benefiting society at large.

In addition to these key features, it is noteworthy that a technology that would not understand how humans function would not be able to benefit them. Hence, behavioral principles should be included, as highlighted recently in a report published by the World Economic Forum: “Contemporary psychology highlights the power of designing systems that go with the grain of human psychology. Attention, construal and motivation are three core psychological processes that can be harnessed to serve as behavioral design principles” [5].

4 A FAIR DATA-DRIVEN PROCESS

The mere definition of fairness does not meet consensus across all the academic fields that have investigated it and tried to address the topic over the past centuries. The strong context-dependence of (the perception of) fairness makes it all the more complicated for all stakeholders in HRM to create or buy fair AI-based solutions.

The Cambridge Dictionary provide the following definition of fairness as a noun: “The quality of treating people equally or in a way that is right or reasonable”. However, the outcome of this very simple search does not take into account the multidimensionality of fairness [2, 44] that cannot be simply explained in the terms provided by this definition.

Mehrabi et al. [24] defines fairness as “the absence of any prejudice or favoritism toward an individual or a group based on their inherent or acquired characteristics”. Of course, there are many other definitions of fairness, but from an operational standpoint, in our everyday lives, fairness is often a perception and a personal appreciation one has of whether a situation or a decision is just, at a given moment in time and in a given social setting [17]. Like beauty — or many other judgments that we make in a daily basis for that matter — fairness lies in the eye (actually the brain) of the beholder. There is therefore a high level of intrapersonal and interpersonal variability when one needs to address whether a decision is fair, for example [22, 41]. The reason is that, like many other things in our lives, fairness appreciation is modulated by a multitude of contextual factors, such as cultural differences, level of education or even gender or age to name only a few [38, 55].

Fairness was a philosophical and sociological matter before it became a technological one. Its study and implementation within the framework of AI is formalized by the definition of a set of rules and equations allowing its quantification and verification. Not including information such as gender, age, or birth country could, in some instances, can contribute to improve fairness of outcomes, in theory. However, there are cases such as classification or association

rules where the exclusion of these bits of information about the candidate have shown not be effective in order to produce fair outcomes [36]. In fact, only a few metrics [8] are particularly used: predictive parity [52], demographic parity and equalized odds [20]. Being able to control for the three aforementioned features turns out to be not only very difficult, but can be counterproductive, as illustrated by Kleinberg et al. [23] who provided statistical evidence for their mutual exclusion in most cases.

4.1 Data Distribution and Bias Analysis

Our mission is to provide access to employment to those who are seeking a job, and to offer them the ability to thrive by learning new hard and soft skills, regardless of their age, gender, origin, education, or level of professional experience. Non-discrimination and limited opportunities to learn are major issues blue collar workers face on a daily basis. To date, there is no satisfactory solution available in order to address this issue either in Europe, the US, or the UK. This is why we decided to have a specific focus on young trains, coaches, and recruits young people who are Neither in Employment nor in Education or Training (NEET) in the retail, logistics, and manufacturing industries. There are unfortunately over 2 million people referred to as NEET in France, 10 million in the United States of America and 1 million in the United Kingdom [34]. We use our technology to vet priority beneficiaries, ensuring a successful first/back to work experience with our client (the future employer) through training, mentoring, mobility and financial services and measure results as positive exists after six months (i.e., the NEET has worked more than sixty days over the period, has signed a contract superior to thirty days, has created their own job or is in training). In 2021, our company is staffing 43% of NEET people on our missions, out of approximately fifteen thousand temporary workers (who work in average 5% more hours than other temporary workers).

Given this context, we want to ensure that no particular group is discriminated by our mathematical models and algorithms. Our proprietary database is constituted of applications made by temporary workers in France who are voluntarily and willingly applying for jobs. The atomic item is composed by the set of attributes related to a temporary worker, the set of information related to the job description to which the candidate could (or would) apply for and a label which describes the outcome of the application (1 if the worker was recruited, 0 otherwise). We also focused on sensitive attributes such as gender, nationality, birth country, need for residence permit (as needed by foreign nationals working on French soil) and education (some workers have declared an education, but some did not).

For example, our hypothesis is that applicants who need a residence permit are more likely to be negatively affected by the model because of a possible bias in our database. The same thing goes for the other sensitive attributes. Therefore, we use dummy variables to categorize what we assume would be a group of candidates that would be “favored” by the algorithm, as opposed to the group that would be “discriminated” one:

- gender: male or female;
- nationality: French nationality or not;
- place of birth: born in France or not;

Table 1: Size of the population for profiles that are recruited versus not recruited, across their belonging to a (assumed to be) favored (Fav.) versus discriminated (Disc.) group

Attributes	Recruited		Not recruited	
	Fav.	Disc.	Fav.	Disc.
gender	71k	31k	70k	32k
nationality	67k	22k	27k	25k
place of birth	60k	24k	17k	16k
education	33k	69k	24k	78k
RP requirement	70k	19k	32k	19k
18-25 years old	29k	60k	19k	33k
25-35 years old	32k	57k	20k	32k
35-45 years old	16k	72k	9k	43k
45-55 years old	8k	81k	3k	49k

- education: has declared an education level or not;
- residence permit (RP) requirement: can work without a residence permit or need to have one;
- 18-25 years old: in the 18-25 age range or not;
- 25-35 years old: in the 25-35 age range or not;
- 35-45 years old: in the 35-45 age range or not;
- 45-55 years old: in the 45-55 age range or not.

We conducted an analysis across these sensitive attributes to assess the fairness of the outcomes provided by our model using the Equal Opportunities framework as defined by the FairLearn toolkit [3]. Before implementing a fair algorithm, we analyzed the data to observe possible biases or under-representation of some categories. Table 1 indicates the size of each population, along with every sensitive attribute, broken down in two outcome categories (recruited versus not recruited) and by the value of each sensitive attribute (assumed to belong to the so-called “favored” group versus the “discriminated” one). For instance, we can see that the label has the same distribution regardless of the gender. Indeed, the gap between the number of recruited males and the number of males that are not recruited is not significantly different from the difference between the number of recruited females and the number of females that are not recruited.

Our goal is to be able to identify unjustified dependencies between outcome group and some sensitive attributes in our dataset. Female candidates, for example, are historically less prone to have an application approved compared to male ones. The same goes with the other attributes previously mentioned. The reader is invited to kindly keep in mind that this first step is not about analyzing the model nor its consequences, but truly to highlight unfairness in the dataset. For the data analysis part, we compute two scores which allow us to appreciate the disparity and the parity of recruitments. We define a sensitive attribute A which indicates the belonging to a group considered as discriminated and $\bar{A}$ the belonging to the privileged group. Considering variable Y to describe the outcome of the worker’s application, we first computed, for each attribute, the Disparate Impact on the data:

$$\text{Disparate Impact} = \frac{P(Y = 1 | A)}{P(Y = 1 | \bar{A})} \quad (1)$$

Table 2: Disparate Impact and Statistical Parity in our Dataset

Attributes	Disparate Impact	Statistical Parity
gender	0.98	-0.01
18-25 years old	1.06	0.04
25-35 years old	1.03	0.02
35-45 years old	0.97	-0.02
45-55 years old	0.87	-0.09
RP requirement	0.72	-0.19
education	0.82	-0.11
nationality	0.66	-0.24
place of birth	0.78	-0.17

This formula expresses a comparison between two probabilities: i) the probability for a member of the discriminated group to have the favorable outcome (to be recruited in our case); ii) the probability for a member of the favored group to have the favorable outcome. The other formula we use, is Statistical Parity:

$$\text{Statistical Parity} = P(Y = 1 | A) - P(Y = 1 | \bar{A}) \quad (2)$$

Using both Disparate Impact and Statistical Parity affords a broader vision and enables us to find effects that can be hidden by the distribution of data.

Given these equations, there is no pre-determined threshold (neither by law nor by regulations) for which an observation is considered biased or not. Feldman et al. [13] argues that a correct Disparate Impact should not be less than 0.8 and not greater than 1.25. These boundaries are based on the “80% rule” recommended by the U.S. Equal Employment Opportunity Commission [50] on uniform guidelines on employee selection procedures in 1979. Regarding the Statistical Parity, without a well-defined threshold in the scientific literature, we consider a gap of -0.1 too large to be unconsidered in the field of recruitment. For instance, this could mean that a man would have 10% more chance to have a validated application. Table 2 presents these two scores for all the sensitive attributes we studied.

We can see there is no clear imbalance when it comes to age or gender. For these variables, Disparate Impact is quite close to 1 and Statistical Parity is close to 0. This means there is no strong under-representation of women among the approved applications compared to men, vice versa. If we consider nationality, for instance, Disparate Impact is particularly low and Statistical Parity is particularly far below zero. This shows that the frequency of approved applications for people who are not French is significantly lower than for French people. We could give the same description for some other variables such as being born in France, education and whether the person does need a residence permit.

Here we clearly see that the distribution of the label is not the same across sensitive attributes. The assumed to be the favored group (people of french nationality) usually benefit from the preferred outcome, while foreign applicants get the preferred outcome roughly half of the time.

From this analysis and examination of our data we did find biases and under-representation that are present for not all but a few of the defined variables (RP requirement, education, nationality, and place

of birth). Our data is based on accurate events and is only meant to describe human behavior. This does not mean a model trained on this data will generate discriminatory rules, since the model is not informed about sensitive attributes. We do think that, this approach represents an appropriate methodology to understand the data. We can also observe the dependency between the label and the sensitive attribute.

4.2 A Set of (Technical) Fairness Guidelines

When it comes to supervised learning tasks in a human resources management context, the Equality of Opportunity metric is considered the more relevant one by the scientific literature [20]. If $\hat{Y}$ is a binary predictor of the outcome of a worker application and Y the associated ground truth, we consider Class 1 as the preferred outcome in the classification task (the worker was recruited). Given a sensitive attribute A indicating the belonging to a group considered as discriminated and $\bar{A}$ the belonging to the favored group, $\hat{Y}$ is considered equal opportunity with respect to sensitive attribute A if:

$$P(\hat{Y} = 1 | Y = 1, A) = P(\hat{Y} = 1 | Y = 1, \bar{A}) \quad (3)$$

This equation could also be written in terms of another statistical metric: the True Positive Rate (TPR). Given the number of true positives (TP) and the number of false negatives (FN), we compute TPR as:

$$TPR = \frac{TP}{TP + FN}$$

Which gives us as new formulation of the equation (3):

$$TPR_{\bar{A}} = TPR_A \quad (3.1)$$

In light of Equation 3.1 we define the True Positive Ratio Parity as:

$$TPRP = |TPR_{\bar{A}} - TPR_A| \quad (4)$$

This particular metric shows a couple of advantages:

- (1) It allows the perfectly accurate solution to be right, whereas the demographic parity metric might describe accurate data as inappropriate. This fact would not make sense in the context of our work, since we are training a model to approximate the ground truth.
- (2) It compares the ability of the model across groups to select relevant profiles.
- (3) It penalizes the model who perform well on the majority group.

The second point above is particularly important in a context of job matching and recruitment. For instance, a hiring agent would rather select too many candidates than too few. If the True Positive Rate for males were really high(er) compared to the True Positive Rate for females, then the model would either select relevant profiles amongst males more accurately than female ones. Or, the model would select irrelevant male profiles more often than it would select irrelevant female profiles. Using this fairness definition is in fact requiring non-discrimination only within the preferred outcome. One can see that this vision of fairness is more than a probabilistic equation because it is linked to an ideology or a philosophical belief. The TPRP is not a normalized concept, and we are conscious of (bias toward what we consider) a meritocratic approach is therefore favored.

Table 3: Evaluation of the performance (F1-score and ROC AUC) and fairness (TPRP) of our worker/job offer matcher model on each sensitive attribute

Attributes	F1-Score	AUC	TPRP
gender	80.29	89.28	3.50
nationality	86.25	88.27	3.53
place of birth	89.45	85.66	4.54
education	80.29	89.28	3.98
RP requirement	86.25	88.27	2.77
18-25 years old	86.30	88.27	0.90
25-35 years old	86.30	88.27	0.70
35-45 years old	86.30	88.27	0.25
45-55 years old	86.30	88.27	0.97

Ideally, we would set a doorstep or a reference value for this metric. An “alarm threshold” to which we will compare our assessments. However, as far as we know, there is no clear recommendation or handbook in the literature on which value to consider. Therefore, we chose ours somewhat arbitrarily, yet based on our own experience. The threshold itself is meant to ensure there is no group who is disproportionately harmed. Such a True Positive Rate Parity value bears the following question “If we consider relevant applications only, how often do we predict correctly those who belong to a group A versus a group B ?”.

4.2.1 Fairness Assessment. Having processed our data, we then move to evaluating the impact of our model. We examined how the model performs based on the Equal Opportunity fairness definition. In order to have a broader view on the model behaves, we have chosen to use a twofold cross validation to reduce randomness. We therefore computed the average True Positive Rate Parity. We also computed the F1-Score and area under the ROC Curve (AUC) to check on the model’s behavior.

As illustrated in Table 3, across every sensitive attribute, our model exhibits fine F1-Score and AUC. Our model does not make undesirable decisions in order to have a fair output. We can also observe that the different TPRP values reach a maximum 4.54% for the attribute defining whether the worker is French or not. As mentioned before, we do not have a clear threshold concerning this metric. This said, a binary vision where a side of a threshold would be right, and the other side wrong is not necessarily relevant. For instance, the issue is more to make sure there is no group that is disproportionately harmed or disadvantaged compared to another one. All TPRP scores obtained in output of our model are below 5%. Having the trade knowledge in hiring and also by interacting with experienced in-house recruiters, we consider this threshold a good baseline for future iteration of the model at the core of our machine learning solution. Thus, each evolution of the model aiming at improving its performance for the matching task will be evaluated a posteriori and should not exceed this threshold of 5% of TPRP for each sensitive attribute.

4.2.2 Unfairness Mitigation. In our approach to developing fair algorithms, we embraced general artificial intelligence fairness and ethics guidelines [19] but we also needed to consider technical

mitigation techniques. Several approaches have been already introduced.

Hardt et al. [20] proposed to achieve equal opportunity by deriving a given predictor. Such an approach bears interesting advantages, including being model insensitive and focusing on optimizing equal opportunity outcomes while keeping an attractive loss function compared to the predictor it is derived from. But it also requires to constantly include the sensitive attribute in every step. This is not ideal for us since we have many missing values in the real world application of our model. Pre-processing interventions were proposed to transform the data in order to reduce correlations [7, 13]. However, these techniques are meant to mitigate any excess in the True Positive Rate Parity and are not the most relevant when considering an Equal Opportunity definition.

Agarwal et al. [1] introduced a promising approach. The exponential gradient algorithm for unfairness reduction appears to be the most relevant one as far as we are concerned. The model performs good results for a classification task. Moreover, this method benefits from requiring the sensitive attribute in the training part only. This approach is particularly attractive in a context where we are concerned by deployment problems and issues.

The problem that remains is that we still need the sensitive attribute in our process. Therefore, implementing and deploying this approach in our machine learning pipelines would mean to train the model on a fragment of our database only. This would not be coherent with one of machine learning fairness main goals: diversity [46]. Sidestepping missing values means reducing diversity because the instances with missing values could describe a particular usage of our product or bear information inherent to groups. Therefore, regardless the fairness definition one chooses, it is important to take steps ahead of the modeling and prediction parts, including diverse opinions in product design and retrieving complete data on a large population coverage. We must have observations on different individuals with different usages as long as some profiles could be discriminated by the product.

5 CONCLUSION

In the context of human resources management (HRM) of blue collars and NEET people, we introduced an approach to research and development of artificial intelligence algorithms that can predict who the best candidate is for a given job offer while controlling for the fairness of our algorithms' outcomes. We did identify biases across the process that led to a differential treatment of individuals (some being favored some being discriminated) as well as biases towards some sensitive information such as owning a resident permit, having a certain level of education and coming from a certain birthplace.

In light of these biases, we have introduced metrics to assess the level of unfairness in our HRM solution that can correct the outcome to make it fair(er). What we did is embedding in our AI-based automation process algorithmic safeguards that flag possible biases (theory) and measured ones (outcomes) in order for the decisions made by our machine learning algorithm in charge of automating the matching between job applicants and job offers to be as fair as possible, or at the very least less unfair.

This is the first step in a series of R&D phases where, regardless of the task to be automated, fairness will be measured, and algorithmic outcomes will be corrected if our safeguard algorithms estimate that fairness is not observed. This not only allows to ensure a higher level of fairness in the outcomes provided by our algorithms, but also to improve transparency and explainability of the decisions (our) algorithms make.

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